a Benchmark construction
ABIDE I resting-state fMRI
PCP · CPAC · band-pass · no GSR
1,112 → 848 subjects (QC, ≥ 100 vols)
393 autism · 455 controls · 19 sites
AAL-116, Pearson r, Fisher z
Connectome x_i (116 × 116)
one brain graph per subject:
6,670 edges = correlations
343 controls; half per site called "patient"
Disease: the known answer
10 regions, ranked on a held-out set;
shared signal in fake patients only,
×1/×2/×4 real effect (ℓ) → 45 true edges
then, on the planted scans
Scanner: the known trap
10 other regions where sites differ most,
sized from real hospital differences;
scanner-B copy of each scan → 45 edges
in each of 10 leave-site-out folds
Shortcut ρ: scanner linked to label
train + val.: scanner B for ρ of patients
and 1 − ρ of controls (ρ = 0.5 → 0.9);
unseen test site: always 50/50
Real-data checks (no planting)
hospital vs autism signal · scanner map
from training data · real autism labels
b Training and evaluation
Fix (one per run)
plain
ComBat ×4
CF-Site
oracle
ComBat: EB yes/no × diagnosis kept yes/no
CF-Site: train on x_i and $\tilde{x}_i$ (other scanner);
oracle: $\tilde{x}_i$ = the true scanner-B scan
train f: fake patients vs fake controls
Model f (one of four)
Log. regression
6,670 edges, L2
expl.: |coef.|
MLP
6,670 → 64 → 2
ReLU, dropout 0.5
GCN (BrainGB)
top-20% graph
2 × GraphConv 16
BrainNetCNN
2 × E2E 32 → E2N 64
N2G 256 → FC → 2
neural: Adam, lr 10^{-3}, 150 epochs;
BrainGNN and BNT fail the sanity test
one score per edge
Group explanation S
saliency (main) · IG · input mask,
averaged over the test site
rank edges by S, score the top 45
Scores
disease AUPRC (chance 0.008)
scanner share of the top 45
accuracy on the unseen test site
per split: 3–6 splits × 3 seeds
Gain or tie
bootstrap CI · sign test · paired t · Holm
gain: CI excludes 0 and Holm p < 0.05